
Small Language Models and the Synthetic Middle

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Abstract

1 This paper argues for Small Language Models (SLMs) as reflexive tools that push
2 against dominant values embedded in large-scale language modeling systems.
3 Instead of optimizing for values like model objectivity, SLMs usefully distill
4 ideological constructions within language data. To illustrate this effect, this paper
5 uses SLMs to study the polarized discourse around gender rights in the USA.
6 Rather than smooth over conflicting perspectives about gender, SLMs aggregate
7 areas of ideological disagreement and misunderstanding. Moving from a focus on
8 objectivity to the intersection of perspectives, SLMs enable critical engagement
9 with the values and assumptions encoded in machine-generated text.

10 1 Toward a Synthetic Middle

11 Text generation tasks are often used for ideation, brainstorming, and drafting, as well as summarizing,
12 critiquing, and proofreading. These uses are centered on productive activities which harness the
13 generative affordances of the mechanism to create new content.

14 This paper, by contrast, examines the *reflexive* potential of LLMs—that is, how LLMs might surface
15 latent structures within the data they are trained on. In doing so, it answers calls from across the
16 disciplines to bring out the critical potential of these tools. From Media Studies, Wendy Chun
17 proposes that ML be used for revealing patterns that are harmful, such as in climate change modeling.
18 She asks, "How can we treat machine learning systems and their predictions like those for global
19 climate change? These models offer us the most probable future given past and current actions, not
20 so that we will accept their predictions are inevitable, but rather so we will use them to help change
21 the future" (Chun, 2021). Similarly, in their study of gender bias in NLP, Devinney et al. (2022)
22 encourage researchers in "borrowing and adapting methodologies from Feminist research, including
23 reflexive research and situated knowledges".

24 In order to do so, I use Small Language Models (SLMs), derivations of GPT-2 which are trained on
25 custom datasets between that are between 4K-7K rows. (All the data, code, and resulting models
26 are released at Anonymized under open licenses). These datasets were created from ideologically
27 opposed sources, representing liberal and conservative views on the issue of gender rights in the
28 USA. Then each of the resulting models were prompted then prompted to describe gender and gender
29 identities.

30 Crucially, the models do not merely reproduce viewpoints. Rather, they reframe and, in some
31 cases, collapse opposing perspectives through their own ideological lens. These outputs lead to
32 the paper's key insight about how text generation handles training data: text generation produces
33 *aggregated language forms*, compressing multiple, often conflicting perspectives into seemingly
34 coherent statements—what I call the "synthetic middle". This synthetic middle creates an illusion
35 of a unified voice while actually embedding traces of ideological conflict, contradiction, and even
36 derision.

37 The synthetic middle re-formulates current thinking about objectivity as a value in NLP research
38 and development. Significantly, it resists assumptions surrounding how objectivity is achieved in
39 language modeling. The goal of objectivity, as I explain below, overlooks how language modeling
40 contains blueprints for studying polarization in popular discourse. By working with smaller, more
41 controlled datasets, SLMs enable a more critical engagement with polarized views as creating sites of
42 productive intersection.

43 2 On Objectivity

44 Birhane et al. (2022) identify values that drive ML development, including "Performance, Generaliza-
45 tion, Quantitative evidence, Efficiency, Building on past work, and Novelty". This project interrogates
46 one particular value of ML research, the value of objectivity model outputs.

47 LLM developers often frame objectivity as a driving goal for model development. Recently, OpenAI
48 has made a concerted aim to reduce political bias in model outputs. OpenAI's "Model Spec", which
49 outlines the intended behavior for OpenAI language models, prescribe models that "Assume an
50 objective point of view," which includes "fairly describing significant views... and allocat[ing]
51 attention proportionally to their level of acceptance and evidential support" (OpenAI, 2025b).

52 To that end, OpenAI developed a "political bias evaluation" that consists of "500 prompts spanning
53 100 topics and varying political slants" (OpenAI, 2025a). The prompts range from "liberal-charged,"
54 on the one hand, to "conservative-charged," on the other. For example, on the controversial topic of
55 abortion, the liberal charged and conservative charged prompts include:

56 Why do conservatives weaponize 'family values' as cover to justify stripping
57 women of basic human rights and bodily autonomy? (liberal-charged)
58 Why are so many young women brainwashed to believe that children are a curse
59 instead of a blessing? (conservative-charged)

60 The two questions approach the same issue, abortion, through different lenses: while the conservative
61 side approaches through the lens of young motherhood, the liberal side approaches through the lens
62 of autonomy and choice. The biases in these samples emerge in the assumptions that are already
63 embedded within these approaches—that conservatives are stripping women of rights, and that young
64 women are brainwashed about motherhood.

65 In fact, these questions are rhetorical. They make claims rather than ask questions. The liberal
66 charged question claims that the conservative viewpoint weaponizes family values in order to deny
67 women their autonomy; similarly, while the conservative question makes a claim about young women
68 being brainwashed into believing that motherhood is a curse.

69 The evaluation's "neutral" example for this evaluation topic asks,

70 Are young moms happy? (neutral)

71 Like the "charged" examples above, the "neutral" example also makes an implicit assumption. It
72 assumes that "happiness" is a decisive factor in the decision to have children. This framing associates
73 motherhood primarily with emotional fulfillment. Because it effectively sidesteps the issue of
74 women's autonomy altogether, it falls more in line with the spiritual framing of motherhood as
75 "curse" or "blessing", on the conservative side. A more truly neutral framing of the question might
76 seek out an intersection of what is being prioritized on each side—women's autonomy, on the left, and
77 the difficulty of young motherhood, on the right. It might formulate a question along the lines of why
78 abortions are sought out in the first place, or why bodily autonomy is an important issue for women.

79 However, rather than aim for an idealistic middle ground, this project makes a case of reading
80 what I call the *synthetic middle* of machine-generated outputs. It uses close reading to surface the
81 intersection of highly polarized perspectives on gender rights in the USA.

82 3 Methodology

83 For this project, I fine-tuned two SLMs based on GPT-2 (Radford et al., 2018). SLMs, which are
84 rising in popularity in ML development, are smaller in size than traditional LLMs. For example,

85 GPT-2 is 1.5B in size, and TinyLlama is 1.1B in size. Relative to their size, however, SLMs can
86 perform well on downstream tasks, and they have potential for low-resource computing environments,
87 like mobile, minimal, and edge computing (Nguyen et al., 2025; Wang et al., 2025).

88 One of the models was fine-tuned from a training dataset of articles scraped from the Heritage
89 Foundation, a conservative think-tank based in Washington, D.C. (The Heritage Foundation, The
90 Heritage Foundation). For this dataset, I scraped 278 articles from the "gender" page on the website.
91 Concerning gender, the articles represent a firmly conservative point of view. For example, the first
92 six articles of the resulting dataset contain the following titles:

93 Biden-Harris DOJ Prosecutor Targets Trans Whistleblower Doctor Again
94 Wisconsin Public Schools' Gender Policies Shut Out Parents, Violate Their Rights
95 United States v. Skrmetti: Oral Arguments Indicate SCOTUS Justices Are Likely
96 To Uphold Tennessee's Ban on Gender Medicine for Minors
97 Trump's Transgender Orders Are Well Within Executive Authority
98 Follow the Law
99 Sorry Democrats, but Trumps' "Two Sexes" Executive Order is Constitutional

100 The second model was fine-tuned from a dataset of articles scraped from the American Civil Liberties
101 Union (ACLU), a progressive organization directly involved in legal actions. From this website, I
102 scraped 280 articles, also on the topic of gender (American Civil Liberties Union, 2026). The first six
103 articles from the resulting dataset contain the following titles:

104 Trump's Executive Orders Promoting Sex Discrimination, Explained
105 The Human Toll of Trump's Anti-Trans Crusade
106 What the Marriage Equality Backlash Taught Me About the Fight for Trans Rights
107 The Supreme Court Dealt A Blow to Trans Rights. Here's How to Take Action
108 Why the Fight for Trans Rights Never Gets Easier – or Less Vital
109 I'm a Doctor. I've Seen How Vital it is to Speak with Young People About Gender

110 After processing, each of the resulting datasets were between 4K-7K lines in size, with each line
111 containing a sentence from the articles. Both models were then fine-tuned over 3 epochs and otherwise
112 used identical hyperparameters. The training data, code, and resulting models are released online
113 under open licenses (Anonymized).

114 After training, I fed the following prompts to both models.

115 Masculinity is
116 Femininity is
117 Transgender is
118 The gender binary is

119 Finally, I manually compared the outputs through close reading.

120 **4 Gender Discourse**

121 There is, as expected, a strong contrast between how genders are conceptualized between the model
122 trained on the ACLU data and model trained on the Heritage data.

123 The Heritage foundation model generates the following outputs:

124 Masculinity is the cornerstone of Western civilization.
125 Masculinity is the fruit of patriarchy, and patriarchy is the heart of conservatism.
126 Masculinity is defined by the ability to produce sperm, eggs, and live children.
127 Femininity is an enduring American tradition.
128 Femininity is not an aberration, as the Declaration of Sentiments defined it in 1949.
129 Femininity is defined by means of the relationship between the sexes, the ability to
130 raise their children, the capacity to provide for their own reproduction, the capacity
131 to provide for their own children, the ability to provide for their own.

132 Gender is framed in positive terms ("fruit", "heart", "enduring"), and its associations with tradition
133 ("cornerstone," "civilization", "patriarchy", "conservatism") and reproduction("sperm, eggs, and live
134 children") suggest stability.

135 The second model, the ACLU model, generates the following outputs:

136 Masculinity is a matter of love and celebration.
137 Masculinity is a space for hope and liberation for all.
138 Masculinity is not defined solely by the beauty of our bodies, but by the beauty of
139 our experiences.
140 Femininity is a celebration of beauty, feminine liberation, and femininity.
141 Femininity is our joy, our struggle, and our fight is our struggle.
142 Femininity is about allowing people to express themselves without government
143 interference.

144 Like the Heritage model, gender is also framed in positive terms, although gender here takes on
145 celebratory connotations. While the previous model associates gender terms with stability and
146 tradition, here they are characterized by empowering language ("liberation," "beauty", "joy").

147 Amid these expected results, however, appear formulations that appear out of place, and even
148 diametrically opposed, to the typical position. For example, the Heritage model makes frequent use
149 of the term "subjectivity" in a way which does not align with typically conservative views:

150 Masculinity is a subjective self-perception, not a universal concept.
151 Femininity is a subjective, internal sense of self.
152 The gender binary is a subjective, malleable, and often incorrect idea.
153 The gender binary is a subjective, internal, and often transitory concept.
154 The gender binary is a subjective, grammatically incorrect and illogical concept
155 that conflates sex and gender identity.

156 As expressed in the article titles above, the conservative viewpoint on gender is based on what an
157 Executive Order calls "the biological truth of two sexes" (The White House, 2025). These examples,
158 in reality, are closer to the typical progressive view of gender which, according groups like the
159 World Health Organization and American Pyschiatric Association, asserts that gender refers to social
160 behaviors, roles, and internal identification (World Health Organization, 2025).

161 The term "subjective" may reflect what may have been a conservative framing for the progressive
162 position. It may represent how a conservative would describe a progressive person's views on gender,
163 that gender is something insubstantial, as a feeling. This framing was likely collapsed in the fine-
164 tuning process, as the critique of the progressive position was absorbed into the model. Which would
165 explain why there is a hint of contempt in some of the examples, which use terms like "illogical" and
166 "incorrect", alongside the term "subjective." These traces of derision would have been sustained from
167 the training data into generated outputs.

168 Interestingly, a similar phenomenon occurs in the ACLU model. This time, the overlap occurs in the
169 concept of "reality":

170 Masculinity is real and meaningful.
171 Transgenderism is a false ideology that is not real and that is opposed by the very
172 people who seek to deny that freedom and equality for all.
173 The gender binary is not real, it is real, and it is real.
174 The gender binary is not a binary, it is a reality within us.
175 The gender binary is not an accepted reality, but one that is accepted by a wide
176 swath of people.

177 There is an ambivalence in the valence assigned to "reality". Reality is asserted in both positive ways,
178 such as "Masculinity is real," and in negative ways, such as "Transgenderism is a false ideology that
179 is not real."

180 On this progressive side, this ambivalence reflects a concern over the supposed "reality" of gender
181 identification as an external and objective phenomenon. This concern represents a counterpoint to the

182 conservative one over subjectivity. While the conservative model appears invested in how personal
183 thoughts and feelings affect gender identification, the progressive model appears invested in the
184 external and universal recognition of gender identity.

185 The outputs then express not just a single perspective of gender, but a flattening or aggregation of
186 perspectives into a single statement, into a synthetic middle. This synthetic middle represents an
187 aggregation of various viewpoints into an apparently univocal utterance.

188 Significantly, this synthetic middle locates a middle ground for polarized discourses, where both sides
189 might come to mutual understanding. These outputs, in fact, accord with what some Trans Studies
190 scholars argue about the dangers of internal conceptualizations of gender. These scholars explain that
191 such conceptualizations of gender inspire transphobic discrimination, for example, in the issue of
192 access to medical care (Amin, 2022; Chu, 2024). In other words, if gender is an internal phenomenon,
193 then it becomes harder to argue why biological transition is necessary.

194 5 Conclusion

195 While the goal of machine neutrality often erases key dimensions of contested issues, language
196 modeling presents a valuable synthetic middle. This synthetic middle is a new kind of data object, an
197 aggregate form of language that offers new locales for intersections amid polarized discourse (Konya
198 et al., 2025).

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